

Fig. 3: Actual-size PCB layout of switch box circuit

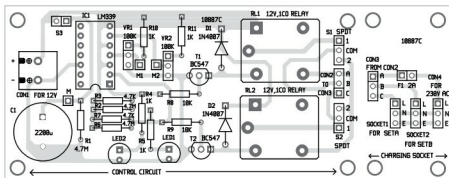


Fig. 4: Component layout for the PCB

the delay circuit. Here, a single R1-C1 circuit produces a common delay time for both the op-amps A1 and A2.

Switch S3 is used to charge capacitor C1 to 12V. When S3 is closed, the voltage at point M is around 12V. When S3 is open, 12V disconnects and

capacitor C1 discharges through resistor R1. Point M is connected to pins 5 and 7 (non-inverting terminals of op-amps A1 and A2) of IC1. Potentiometers VR1 and VR2 are connected to pins 4 and 6 (inverting terminals of op-amps A1 and A2) of IC1. Resistors R2 and R6 are used to provide feedback and stability. Resistors R3 and R7 are used as pull-up resistors. Pins 3 and 12 of IC1 are connected to 12V and ground, respectively.

LED1 and LED2 display the relay status. When S3 is open, the switch box operates normally and the delay circuit remains disabled. When S3 is closed and then opened, the capacitor discharges from 12V to 0V as the delay circuit is enabled.

The voltage at C1 (at point M) is given by the following relationship:

$$V_{C1} = V_{dc} \times e^{-t/(R1C1)}$$

where V_{dc} is 12V.

When the voltage is set at M1 or M2 using the potentiometer (VR1 or VR2), at the same time the capacitor also discharges slowly from 12V to 0V.

If voltage at M > M1 or M2, op-amp outputs are high and relays energise, and vice versa. For example, if you want to de-energise relay RL1 after

3600 seconds (one hour), set the voltage at M1 to 8.47V. After 3600 seconds, the relay will de-energise, cutting off the voltage to Set A and Set B.

The table shows voltage values at point M1, M2 and delay time for different values of R1 and C1.

Switches S1 and S2 connect the line supply directly or through relays depending on their position. When switches S1 and S2 are in position 1, Set A and Set B get power supply through relays. When switch S1 and S2 are in position 2, Set A and Set B get power supply directly without relays.

During normal operation, switch S3 is open. Keep switches S1 and S2 in position 2 to provide line supply directly to Set A and Set B.

Construction and testing

A combined PCB for the switch box circuit (including control and charging sections) is shown in Fig. 3 and its components layout in Fig. 4. You can easily separate the two PCBs.

Enclose them in suitable boxes. Connect respective connectors between the control circuit and the charging socket.

The circuit operates off 12V SMPS. A 3-pin plug with 2m long 3-core wire is used to connect the line supply (L, N and E) to the switch gang box.

Variable voltages obtained through the middle pin of potentiometers provide different delays. Measure the voltage and calibrate the dials of VR1 and VR2 in terms of delay time. Accuracy of time depends on the tolerance of R1 and C1.

Connect CON2 and CON3 using male and female connectors, respectively. Connect 230V AC, 50Hz at CON4.

EFY notes. Since the assembled PCB is placed in the switch gang box, take care between AC and DC wiring. Use an appropriate fuse. Also take care while connecting the relays. Use a 2-mega-ohm R1 in place of 4.7M if you want a different value as shown in the table. **EFY**



K. Murali Krishna is a junior engineer (telecom) in BSNL, Rajahmundry, Andhra Pradesh